Chainesh Gautam

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Education

International Institute of Information Technology, Bangalore **Graduated 2026**
Master of Science by Research (Stochastic Control Lab, Data Science and AI) *3.67/4.0*

Lovely Professional University, Punjab **Graduated 2023**
Bachelor of Technology in Computer Science (School of Computer Science) *7.93/10.0*

Research Experience

Indian Institute of Science Bangalore (Vivitsu Lab) **March 2026 – July 2026**
Research Associate (Advisor: Prof. Gugan Thoppe) *Bangalore, Karnataka*
- Analyzed Reliable Policy Iteration and its Bellman-feasibility and monotonic-improvement properties for mitigating instability in approximate policy iteration with linear function approximation.
- Built a scalable sampled RPI method using Kaczmarz–Motzkin projections and benchmarked it against AMPI-Q, CPI, TRPO, and SAC across multiple control environments like inventory control and Tetris.

Stochastic Control Lab **Aug 2023 – July 2026**
Research Scholar (Advisor: Prof. Raghuram Bharadwaj) *Bangalore, Karnataka*
- Developed a **Gaussian Mixture Q-Network** for uncertainty-aware value estimation, introducing stable responsibility computations and mixture-weight updates, and evaluated its performance against DQN and MoE-based baselines.
- Proposed a CLIP-inspired transfer learning framework that aligns natural language instructions with RL policy weights, creating a shared embedding space to retrieve and transfer expert behaviors based on semantic task descriptions.

IBM Research, IN **May 2025 – July 2025**
AI Research Intern *Bangalore, Karnataka*
- Built an offline-RL pipeline in Gym environments to generate diverse, multimodal trajectories via return-based binning, rejection sampling supporting both single and multi-objective reward structures in D4RL dataset format.
- Trained and evaluated Decision Transformer and Elastic Decision Transformer models, enabling precise return conditioning, controlled policy evaluation, and extensions to multi-objective learning and Pareto-front analysis.

Centre for Artificial Intelligence and Robotics (CAIR), DRDO **Sept 2022 – May 2023**
Research Intern *Bangalore, Karnataka*
- Developed a benchmark dataset of drone imagery captured at day and then converted to low light imagery, enabling standardized evaluation of object detection and tracking algorithms for defense applications used in the military.
- Integrated the SiamRPN++ tracker with TLD (OpenCV) and low-light image enhancement models (e.g. EnlightenGAN), substantially improving tracking accuracy beating state of the art results under extreme illumination constraints.

Publications and Submissions

Accepted / Presented

1. **C. Gautam**, R. B. Diddigi, C. Kamanchi, S. Mukherjee, P. Dayama, S. Kameshwaran. "Trust Guided Decision Transformer", Accepted at **Neural Information Processing Systems (Neurips) 2026** , main track. *arXiv:* http://arxiv.org/abs/2609.31586
2. **C. Gautam**, C. Kamanchi, S. Kameshwaran, S. Mukherjee, P. Dayama, R. B. Diddigi. "Context-Aware Decision Transformer." Accepted at Reinforcement Learning in Big Worlds Workshop at **RLC 2026**. *arXiv:* https://openreview.net/pdf?id=YvY0f8aVJP
3. P. P. Kambhampati, †**C. Gautam**, J. Palaniswamy, M. Rao. "NeuRehab: A Reinforcement Learning and Spiking Neural Network-Based Rehab Automation Framework." Accepted at **Neuromorphic Computing and Engineering Journal**, July 2026. *arXiv:* arxiv.org/abs/2512.17841
4. **C. Gautam**, C. Kamanchi, R. B. Diddigi. "Enhanced Deep Q-Learning with Gaussian Mixtures." Extended abstract accepted at the International Conference on Autonomous Agents and Multiagent Systems (**AAMAS 2026**), Main Track. *https://dl.acm.org/doi/abs/10.65109/KVOL6969*
5. **C. Gautam**, C. Kamanchi, S. Kameshwaran, S. Mukherjee, P. Dayama, R. B. Diddigi. "Insights on Decision Transformers for Multi-Objective Offline Control." Extended abstract presented at the **IBM Research Symposium 2025** (***Best Paper Runner-up Award***)
6. **C. Gautam**, R. B. Diddigi. "CLIP-RL: Aligning Language and Policy Representations for Task Transfer in Reinforcement Learning." Poster at the Multi-Disciplinary Conference on Reinforcement Learning and Decision Making (**RLDM 2025**). *arXiv:* arxiv.org/abs/2512.01616

Personal Projects

Context Robustness in Time-Series Foundation Models **July 2026**

- Implemented a context contamination framework showing that TimesFM, TTM and Chronos rely mainly on the **last 10% of their input history**. Used this to split each series into trend, seasonal and leftover parts and built an ensemble model that gives each part to the model that forecasts it best.
- The ensemble model cut **forecast error by about** ∼**3%** over the best single model on low-seasonality data, and matched it on strongly seasonal data.

Parameter Scaling in RL using Mixture of Experts (Pytorch) **November 2025**

- Implemented a Soft-Mixture-of-Experts DQN with dynamic gating and top-k routing, enabling adaptive expert specialization and balanced expert utilization.
- Benchmarked on Atari-100k, achieving **+15% faster convergence** and ∼**35% lower variance** in returns compared to DQN, demonstrating parameter-scaling benefits without extra compute cost.

Rehabilitation gym environment † **(Pytorch, Gymnasium)** **August 2025**

- Built RL gym environments end to end for upper-limb rehabilitation by modelling patient–robot interaction dynamics, and designed clinically meaningful reward structures, enabling controlled training of assistive movements.
- Evaluated RL algorithm (SAC) to study stability, safety, and adaptability, providing insights into how RL can support personalized rehabilitation protocols. This work was accepted at Neuromorphic Computing and Engineering Journal.

Mentorship

Teaching Assistant **August 2024 – May 2025**
Machine Learning, Reinforcement Learning *IIIT Bangalore*

- Led interactive teaching sessions and practical exercises on machine learning fundamentals, enhancing students practical implementation skills through hands-on programming guidance.

Courses

- Mathematics for Machine Learning
- Reinforcement Learning
- Self-Supervised Learning
- Bayesian Methods and Probabilistic Graphical Models

Skills

Languages Python, C

Technologies Pytorch, Gymnasium, Git, OpenCV

Academic Service

Sub-Reviewer, AAAI 2027 **Montreal, Canada**
Sub-Reviewer, Neurips 2026 **Sydney, Australia**
Sub-Reviewer, ICML MoFA Workshop 2025 **Vancouver, Canada**
RISE (Organizing committee) **2024 and 2025, IIIT Bangalore**
ICVGIP'24 Conference (Head Volunteer) **December 2024, IIITB Bangalore**
Cods-Comad'24 Conference (Volunteer) **January 2024, IIITB Bangalore**